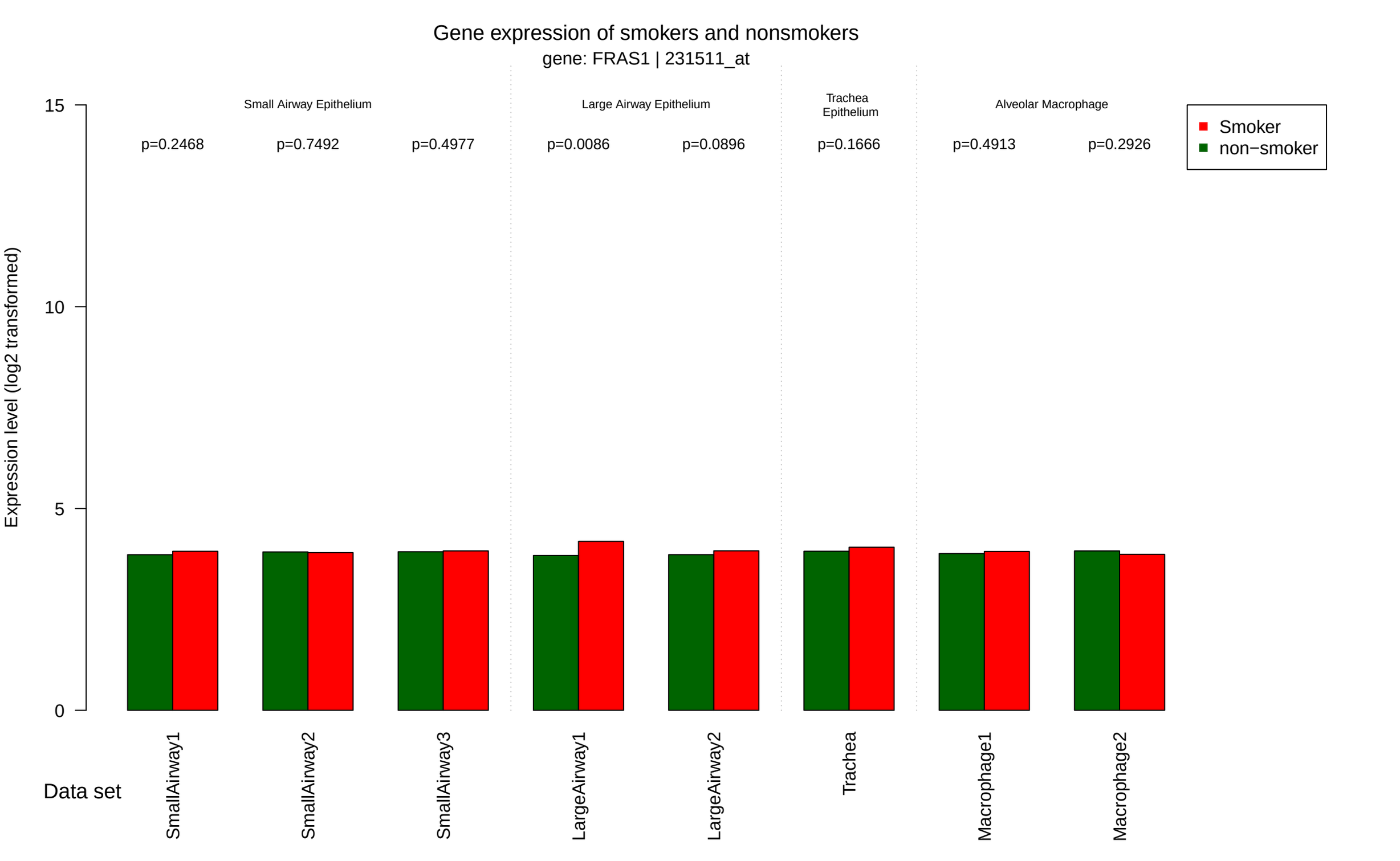
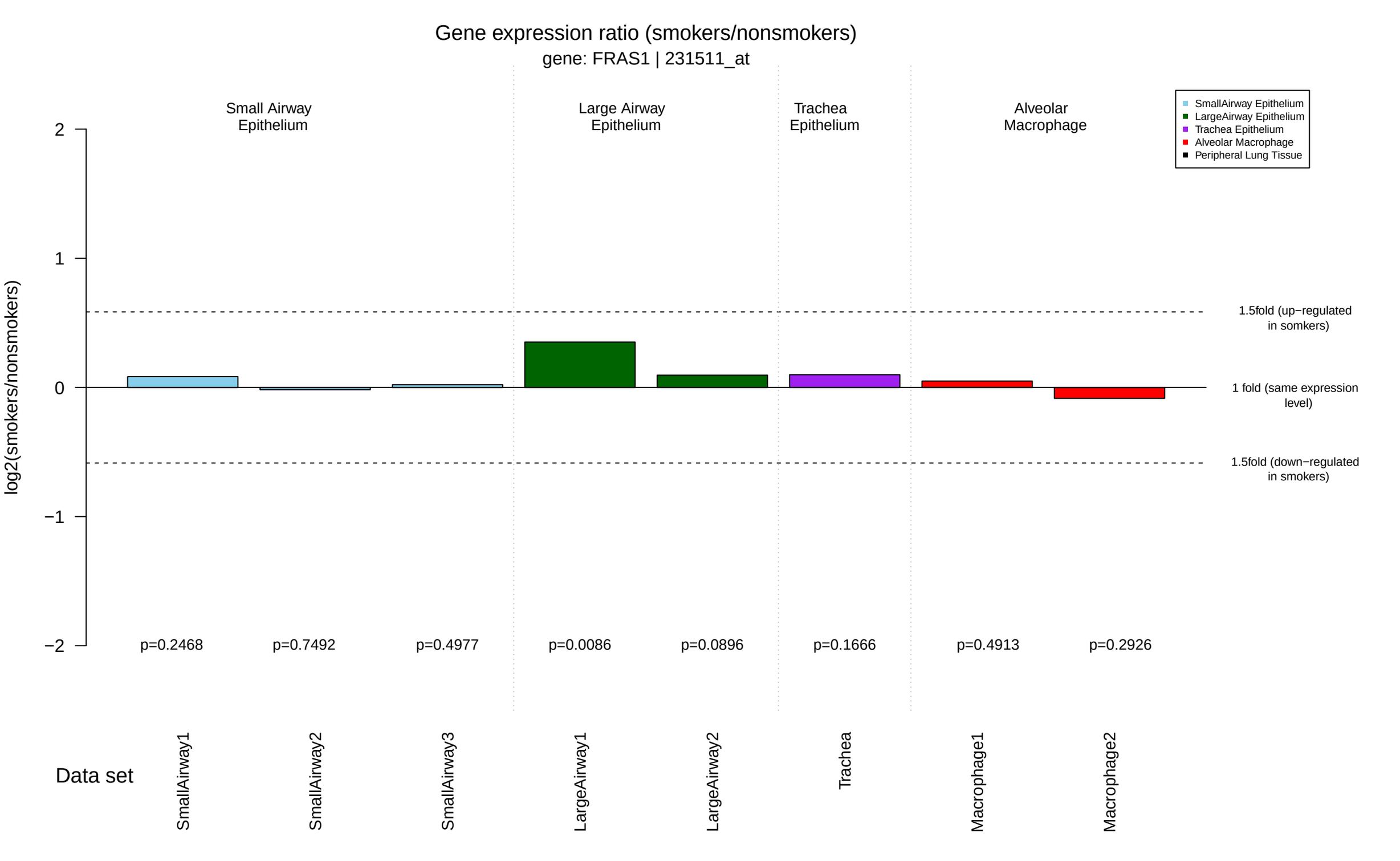
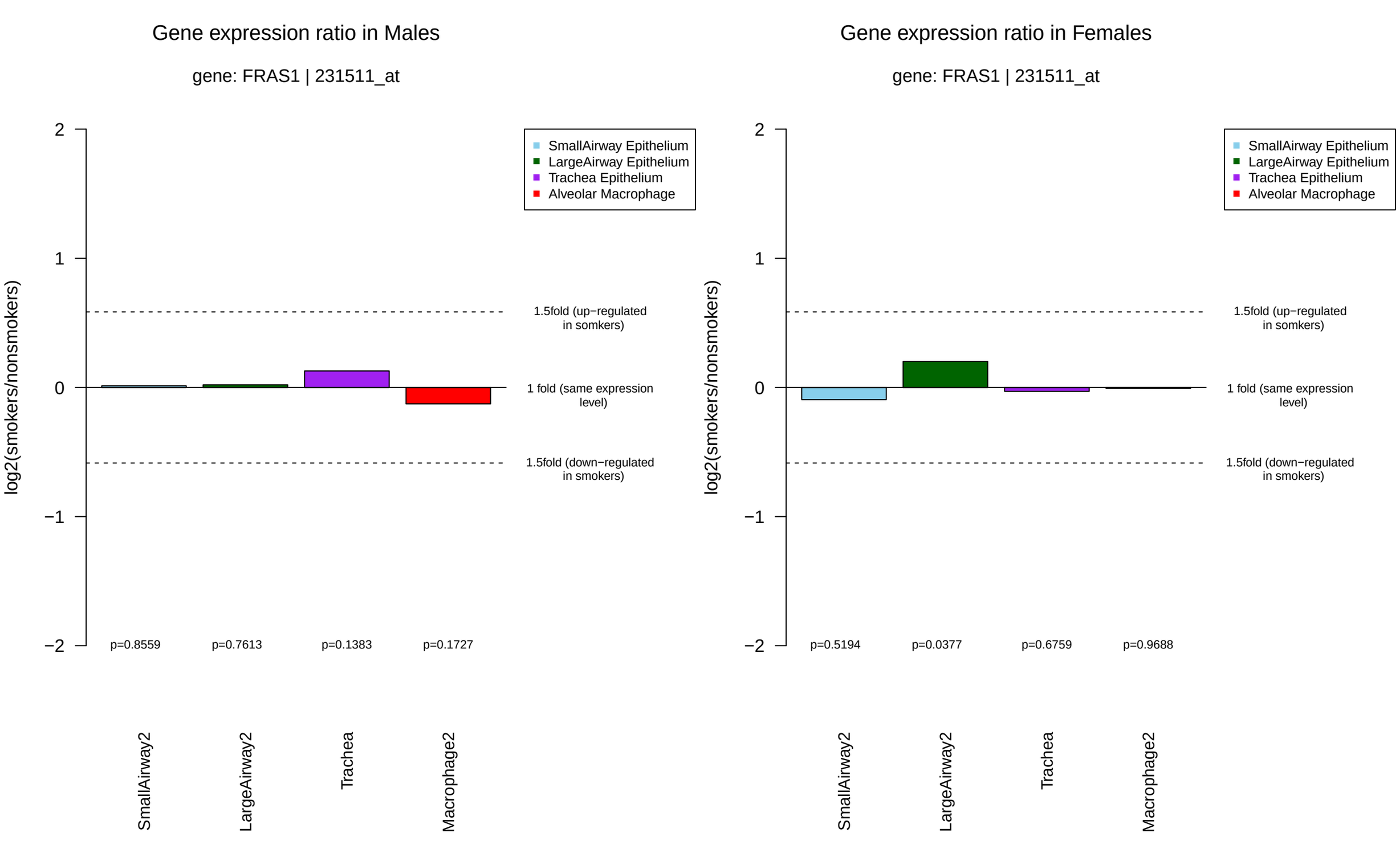
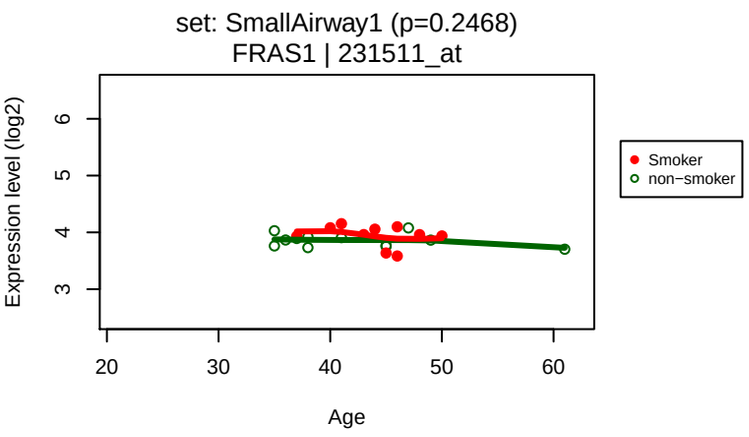
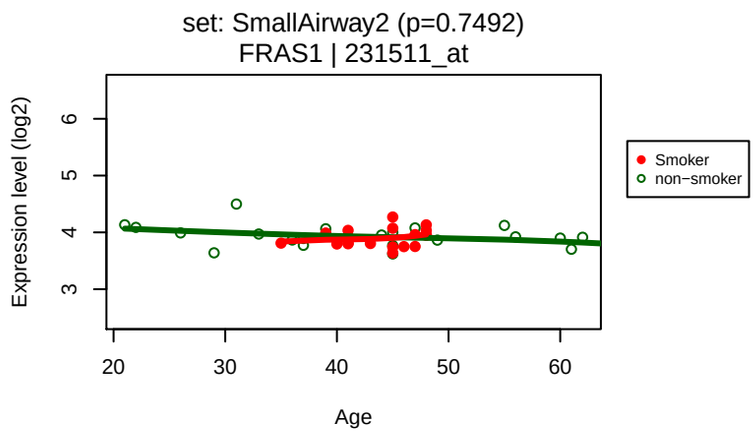
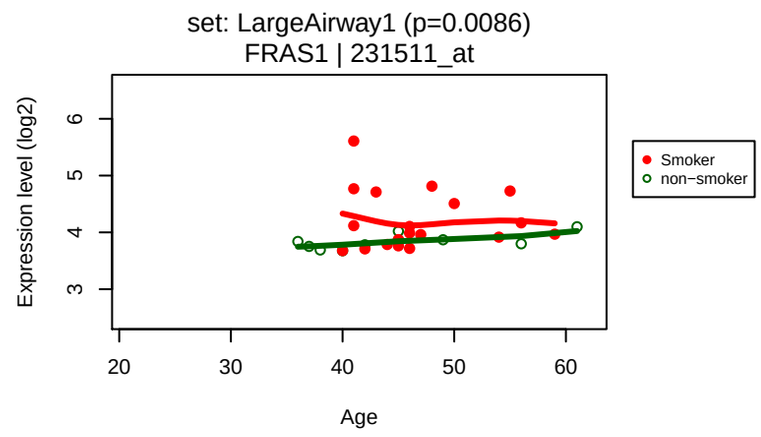

Gene expression of smokers and nonsmokers
gene: FRAS1 | 231511_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.2468
p=0.7492
p=0.4977
p=0.0086
p=0.0896
p=0.1666
p=0.4913
p=0.2926
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

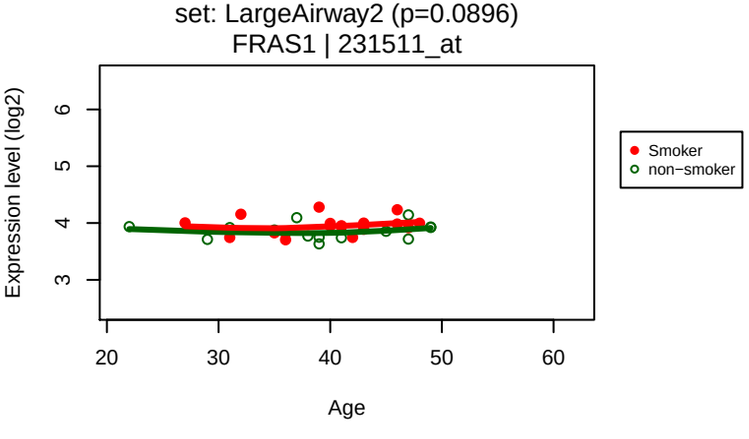

Gene expression ratio (smokers/nonsmokers)
gene: FRAS1 | 231511_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2468
p=0.7492
p=0.4977
p=0.0086
p=0.0896
p=0.1666
p=0.4913
p=0.2926
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

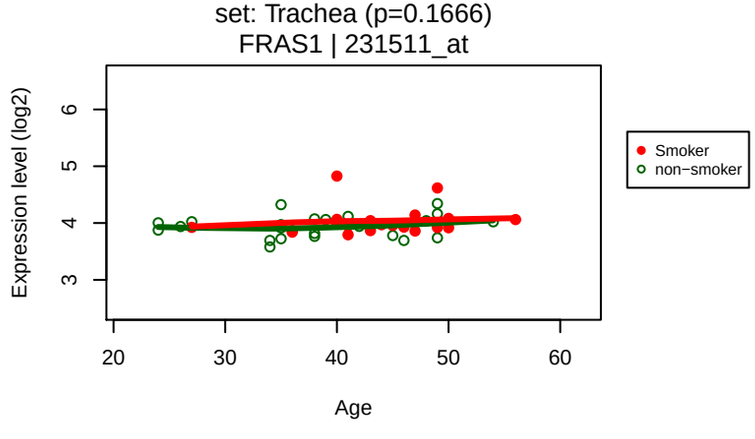

Gene expression ratio in Males
gene: FRAS1 | 231511_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8559
p=0.7613
p=0.1383
p=0.1727
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: FRAS1 | 231511_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5194
p=0.0377
p=0.6759
p=0.9688
SmallAirway2
LargeAirway2
Trachea
Macrophage2

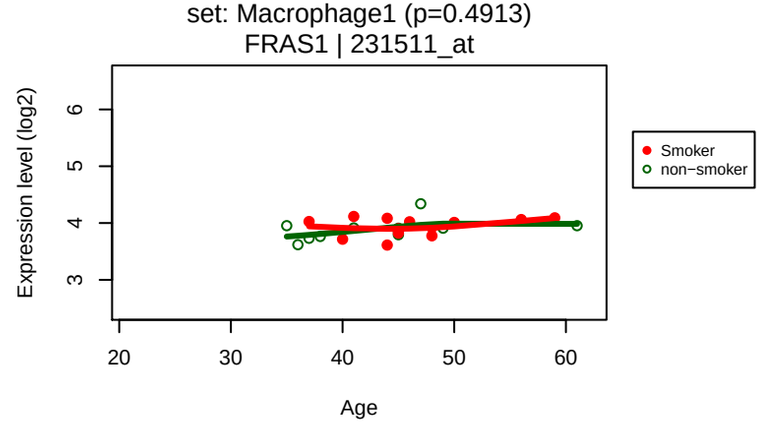

set: SmallAirway1 (p=0.2468)
FRAS1 | 231511_at
Expression level (log2)
Age
Smoker
non−smoker

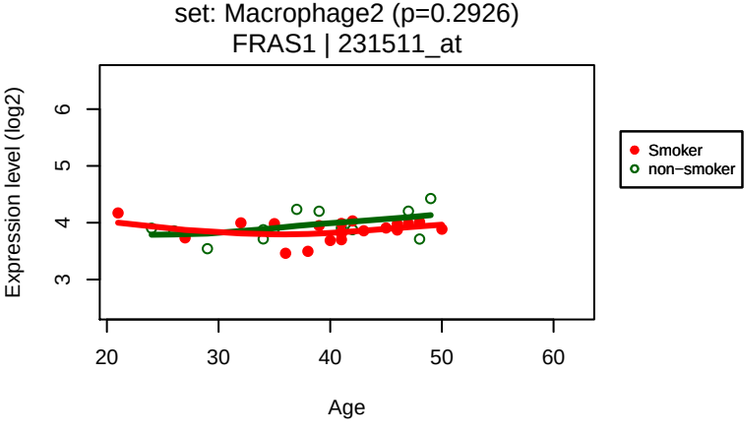

set: SmallAirway2 (p=0.7492)
FRAS1 | 231511_at
Expression level (log2)
Age
Smoker
non−smoker

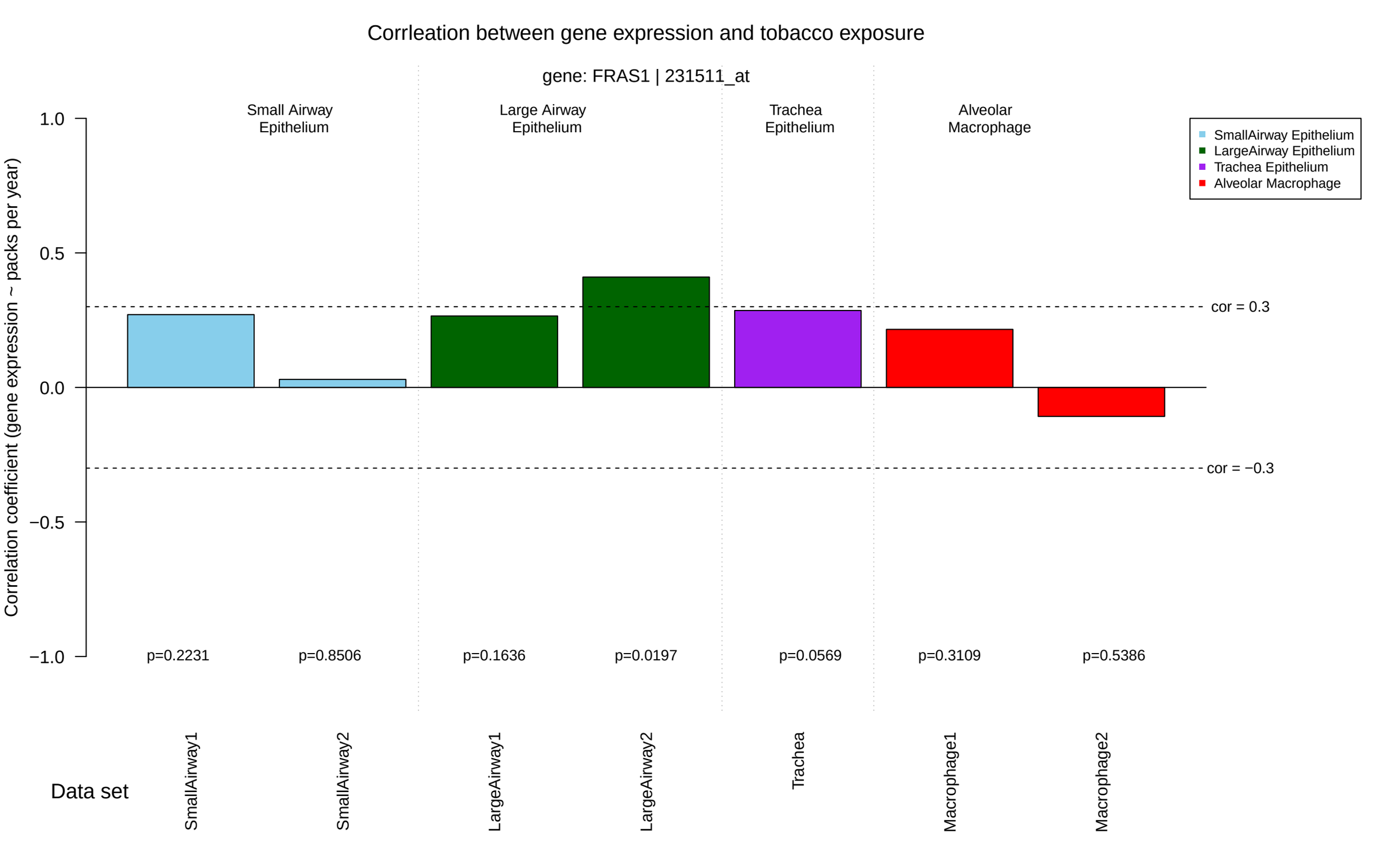

set: LargeAirway1 (p=0.0086)
FRAS1 | 231511_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0896)
FRAS1 | 231511_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.1666)
FRAS1 | 231511_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.4913)
FRAS1 | 231511_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.2926)
FRAS1 | 231511_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: FRAS1 | 231511_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.2231
p=0.8506
p=0.1636
p=0.0197
p=0.0569
p=0.3109
p=0.5386
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2